LAB # 14

MUTABILITY AND IMMUTABILITY

OBJECTIVE:

Understanding and implementing the concept of mutability and immutability.

LAB TASK:

1. Apply concept of mutability and immutability for the task promoted and failed students. The roll number, subject code, and subject name would have to be entered at time of object creation and with getter method these variables should be printed. (Hint: Those students who are failed in previous semester will be registered in immutable class, and promoted students are registered in mutable class)

```java
final class FailedStudent {
    private final int rollNo;
    private final String subjectCode;
    private final String subjectName;
    public FailedStudent(int rollNo, String subjectCode, String subjectName) {
        this.rollNo = rollNo;
        this.subjectCode = subjectCode;
        this.subjectName = subjectName;
    }
    public int getRollNo() {
        return rollNo;
    }
    public String getSubjectCode() {
        return subjectCode;
    }
    public String getSubjectName() {
        return subjectName;
```

```java
class PromotedStudent {
    private int rollNo;
    private String subjectCode;
    private String subjectName;
    public PromotedStudent(int rollNo, String subjectCode, String subjectName) {
        this.rollNo = rollNo;
        this.subjectCode = subjectCode;
        this.subjectName = subjectName;
    }
    public int getRollNo() {
        return rollNo;
    }
    public String getSubjectCode() {
        return subjectCode;
    }
    public String getSubjectName() {
        return subjectName;
    }
    public void setRollNo(int rollNo) {
        this.rollNo = rollNo;
    }
    public void setSubjectCode(String subjectCode) {
        this.subjectCode = subjectCode; }
    public void setSubjectName(String subjectName) {
        this.subjectName = subjectName;
```

```java
        FailedStudent fs = new FailedStudent(101, "CS101", "Programming");
        System.out.println("2023F-BSE-019");
        System.out.println("Failed Student:");
        System.out.println("Roll No: " + fs.getRollNo());
        System.out.println("Subject Code: " + fs.getSubjectCode());
        System.out.println("Subject Name: " + fs.getSubjectName());
        System.out.println("----------------------------");
        // Promoted Student (Mutable)
        PromotedStudent ps = new PromotedStudent(202, "CS102", "OOP");
        System.out.println("Promoted Student Before Update:");
        System.out.println(ps.getRollNo());
        System.out.println(ps.getSubjectCode());
        System.out.println(ps.getSubjectName());
        // Updating values
        ps.setSubjectName("Data Structures");
        System.out.println("Promoted Student After Update:");
        System.out.println(ps.getSubjectName());
```

utput - FailedStudent (run)

```
Failed Student:
Roll No: 101
Subject Code: CS101
Subject Name: Programming
----------------------------
Promoted Student Before Update:
202
CS102
OOP
Promoted Student After Update:
Data Structures
```

2. Write a program that will calculate the below 4 formulas. Decide what to make mutable and what to make immutable and perform task operations. Formulas are:

Circumference of circle: $\mathbf{C = 2 \pi r}$
Area of circle: $\mathbf{A = \pi r^2}$
Volume of sphere: $\mathbf{V = 4/3 \pi r^3}$
Surface area of sphere: $\mathbf{A = 4 \pi r^2}$
(Hint: Value of pie would be constant and value of radius should be variant)

```java
final class Constants {
    public static final double PI = 3.14159;
}
class Geometry {
    private double radius;   // mutable
    public Geometry(double radius) {
        this.radius = radius;
    }
    public void setRadius(double radius) {
        this.radius = radius;
    }
    public double getRadius() {
        return radius;
    }
    public double circumferenceOfCircle() {
        return 2 * Constants.PI * radius;
    }
    public double areaOfCircle() {
        return Constants.PI * radius * radius;
    }
    public double volumeOfSphere() {
        return (4.0 / 3.0) * Constants.PI * radius * radius * radius;
    }
    public double surfaceAreaOfSphere() {
        return 4 * Constants.PI * radius * radius;
    }
}
public class Lab14_Task2 {
    public static void main(String[] args) {
        Geometry g = new Geometry(7);
        System.out.println("2023F-BSE-015");
        System.out.println("Radius: " + g.getRadius());
        System.out.println("Circumference: " + g.circumferenceOfCircle());
        System.out.println("Area of Circle: " + g.areaOfCircle());
        System.out.println("Volume of Sphere: " + g.volumeOfSphere());
        System.out.println("Surface Area of Sphere: " + g.surfaceAreaOfSphere());
        // Changing radius (mutable)
        g.setRadius(10);
        System.out.println("\nAfter Changing Radius:");
        System.out.println("Radius: " + g.getRadius());
        System.out.println("Area of Circle: " + g.areaOfCircle());
```

Output - Constantss (run)

```
run:
2023F-BSE-015
Radius: 7.0
Circumference: 43.98226
Area of Circle: 153.93791
Volume of Sphere: 1436.7538266666666
Surface Area of Sphere: 615.75164

After Changing Radius:
Radius: 10.0
Area of Circle: 314.159
```